

Introduction to Clifford algebra applications: Quantum eigens and the Inverse of a multivector

Arturas Acus** and Adolfas Dargys*

*Center for Physical Sciences and Technology, Semiconductor Physics Institute,
Saulėtekio 3, Vilnius, Lithuania

**Institute of Theoretical Physics and Astronomy, Vilnius University, Saulėtekio 3,
Vilnius, Lithuania

January, 2019

1 Introduction to geometric algebra

2 $Cl_{3,0}$ quantum mechanics

- Standard quantum mechanics
- How to dispense with matrices?
- Two level model
- Example of monolayer graphene
- The algorithm

3 Multivector inverse for $n = p + q \leq 6$

- Introduction
- Grade-negated self-product
- Inverse of general MV of algebras of dimension $n = p + q \leq 5$
- Inverse of general MV of algebras of dimension $n = p + q = 6$

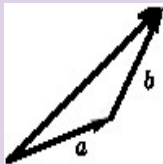
4 Coordinate-free solution of quaternionic Sylvester equation

5 Conclusions

Introduction to geometric algebra (GA)

A little provocation in order to attract the attention.

A question



Polar vector + axial vector = ?

Introduction to geometric algebra (GA)

Can two vectors $\vec{a}$ and $\vec{b}$ always be summed?

- $\vec{a} + \vec{b} = \vec{c}$. For this purpose consider vector reflections.
- What if one of vectors represents a momentum $\vec{a} = \vec{p}$ and the other represents an angular momentum $\vec{b} = \vec{L} = \vec{r} \times \vec{p}$?
(Alternative: electric and magnetic field)
- The standard vector model (Gibbs-Heaviside) is misleading, since it introduces two kinds of vectors (polar/axial).
- Angular momentum, in fact, is not a vector. It is part of plane, which may be characterized by normal vector $\vec{L}$.

Introduction to geometric algebra (GA)

"...for I cannot believe that anything so ugly as the multiplication of matrices is an essential part of the scheme of nature".

sir Arthur Eddington

Vector algebra

- Coordinate free
- Has geometric meaning
- Works only in Euclidian 3D space+inconsistent

Tensor method

- Coordinate method
- No geometric meaning
- Works in all dimensions and space metrics

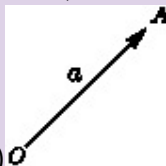
Geometric algebra

- Coordinate free
- Has geometric meaning
- Works in all dimensions and space metrics

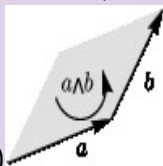
Introduction to geometric algebra (GA)

Geometric algebra objects

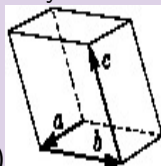
GA deals with oriented objects: External orientation in nD space (called attitude) and internal orientation. Examples: scalar, vectors, bivectors, trivectors, etc..., totally 2^n elements



a)



b)



c)

- a) Translation vector $\mathbf{a}$ is an oriented line segment, which size is $|\mathbf{a}| = OA$. What about radius vectors?
- b) Bivector $\mathbf{a} \wedge \mathbf{b}$ is oriented part of plane, size is an area, shape is not important.
- c) Trivector (pseudoscalar) $\mathbf{a} \wedge \mathbf{b} \wedge \mathbf{c}$ is an oriented 3D volume, orientation is defined by left/right coordinate system made of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, size is the volume, shape is not important

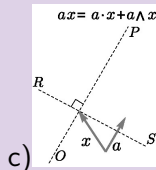
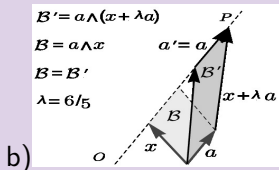
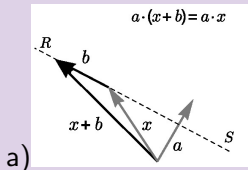
Introduction to geometric algebra (GA)

How to build the oriented objects from vectors? Define three products.

Geometric algebra products

GA has just **one true product** (geometric). The other two ("inner" and "outer") are used for geometric interpretation mainly.

- outer product ($\wedge$), fully antisymmetric, associative [vector product is not!, e.g. $(a \times b) \times c \neq a \times (b \times c)$], however $|a \wedge b| \leq |a| |b|$.
- inner product ($\cdot$), analog of vector scalar product (few types)
- geometric product (space), associative, includes information from both products above, preserves norm $|ab| = |a| |b|$ (very usefull).



Allows **inverse** of ∇a , whereas individual $\text{div } a = \nabla \cdot a$ and $\text{curl } a = \nabla \times a$ do not have have this property!

Introduction to geometric algebra (GA)

Associative algebra = Summation + associative GA product

Mathematica example

Geometric (Clifford) algebra $\iff$ geometric product

$$\begin{aligned} \mathbf{a}\mathbf{b} &= \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \wedge \mathbf{b}, \\ \mathbf{b}\mathbf{a} &= \mathbf{a} \cdot \mathbf{b} - \mathbf{a} \wedge \mathbf{b}. \end{aligned} \quad \text{GEOMETRIC PRODUCT (of vectors)}$$

Solving for product terms results in

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= \frac{1}{2}(\mathbf{a}\mathbf{b} + \mathbf{b}\mathbf{a}), \\ \mathbf{a} \wedge \mathbf{b} &= \frac{1}{2}(\mathbf{a}\mathbf{b} - \mathbf{b}\mathbf{a}). \end{aligned}$$

If vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel, then $\mathbf{a} \wedge \mathbf{b} = 0$. Therefore $\mathbf{a}\mathbf{b} = \mathbf{b}\mathbf{a}$ if $\mathbf{a} \parallel \mathbf{b}$.

Taking $\mathbf{a} = \mathbf{b}$ we have $\mathbf{a} \cdot \mathbf{a} = \mathbf{a}^2 = |\mathbf{a}|^2$. Whereas $\mathbf{a} \wedge \mathbf{a} = 0$.

If vectors perpendicular then $\mathbf{a} \cdot \mathbf{b} = 0$. Therefore $\mathbf{a}\mathbf{b} = -\mathbf{b}\mathbf{a}$ if $\mathbf{a} \perp \mathbf{b}$.

Conclusion:

- Vectors commute $\iff$ they are parallel.
- Vectors anticommute $\iff$ they are perpendicular.

Introduction to geometric algebra

Basis vectors $\mathbf{e}_i$ of GA $Cl_{p,q}$, where (p, q) is the metric

Basis vectors are of **special** importance. They satisfy:

$$\mathbf{e}_i \mathbf{e}_j + \mathbf{e}_j \mathbf{e}_i = \pm 2\delta_{ij}, \quad i, j = 1, 2, \dots, p + q = n.$$

Sign $\mathbf{e}_i^2 = +1$ for $i \leq p$ and $\mathbf{e}_i^2 = -1$ for $p < i \leq p + q$.

For example, for $Cl_{3,0}$ $\mathbf{e}_1^2 = \mathbf{e}_2^2 = \mathbf{e}_3^2 = 1$, signature $(+++)$.

For Minkowski space $Cl_{1,3}$, $\mathbf{e}_1^2 = 1, \mathbf{e}_2^2 = \mathbf{e}_3^2 = \mathbf{e}_4^2 = -1$. Algebras $Cl_{1,3} \not\cong Cl_{3,1}$ (!) are not isomorphic.

Basis vectors are orthonormal, $\mathbf{e}_i \cdot \mathbf{e}_j = 0$, if $i \neq j$. Therefore their outer product coincides with geometric product

$\mathbf{e}_i \mathbf{e}_j = \mathbf{e}_i \wedge \mathbf{e}_j = -\mathbf{e}_j \wedge \mathbf{e}_i = -\mathbf{e}_j \mathbf{e}_i$. In $Cl_{3,0}$ one can make only 3 independent pairs, i. e., $\mathbf{e}_1 \wedge \mathbf{e}_2$, $\mathbf{e}_2 \wedge \mathbf{e}_3$ and $\mathbf{e}_3 \wedge \mathbf{e}_1$. They square to

$$(\mathbf{e}_1 \wedge \mathbf{e}_2)^2 = (\mathbf{e}_2 \wedge \mathbf{e}_3)^2 = (\mathbf{e}_3 \wedge \mathbf{e}_1)^2 = -1.$$

Introduction to geometric algebra

Mathematica example

Matrix representations for $Cl_{2,0}$

$$\mathbf{e}_i \mathbf{e}_j + \mathbf{e}_j \mathbf{e}_i = +2\delta_{ij}, \quad i, j = 1, 2, \dots, p + q.$$

$$\mathbf{e}_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Squares gives identity matrix = scalar element. Multiplication of above matrices gives $\mathbf{e}_{12}$ element

$$1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{e}_{12} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Matrix representations has **no grade** (scalar, vector, bivector) information. Note about $Cl_{3,0}$ invariants: energy (scalar), momentum (vector), angular momentum (bivector), ...

Introduction to geometric algebra

Matrix representations for $Cl_{1,1}$

$$\mathbf{e}_i \mathbf{e}_j + \mathbf{e}_j \mathbf{e}_i = \pm 2\delta_{ij}, \quad i, j = 1, 2, \dots, p + q.$$

$$1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{e}_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad \mathbf{e}_{12} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

$\mathbf{e}_1^2 = +1$ whereas $\mathbf{e}_2^2 = -1$. The most general element $Cl_{1,1}$ can be written as

$$a_0 + a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + a_3 \mathbf{e}_{12} = \begin{bmatrix} a_0 + a_1 & a_2 + a_3 \\ -a_2 + a_3 & a_0 - a_1 \end{bmatrix}.$$

Introduction to geometric algebra

Matrix representations for $Cl_{0,2}$ (quaternions)

$$\mathbf{e}_i \mathbf{e}_j + \mathbf{e}_j \mathbf{e}_i = -2\delta_{ij}, \quad i, j = 1, 2, \dots, p + q.$$

$$q = \begin{bmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{bmatrix}.$$

Taking one of q_i equal 1 or $i = \sqrt{-1}$, and remaining 0's, we find **complex** matrix representations of quaternion basis elements

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}.$$

After multiplication by $-i$, they coincide with Pauli matrices $\hat{\sigma}_z$, $\hat{\sigma}_y$ and $\hat{\sigma}_x$.

For real representations of quaternions we need 4×4 matrices.

Introduction to geometric algebra

Trigonometric expression for inner product

It is useful in basis-free calculations.

In $Cl_{2,0}$ for inner product

$$a \cdot b = |a||b| \cos \varphi,$$

φ is an angle from a to b in anticlockwise direction,

$|a| = \sqrt{a_1^2 + a_2^2}$ is vector norm (length). Can be calculated as

$|a| = \sqrt{a\tilde{a}}$, Tilde means the reversion operation,

$$\widetilde{abc} = cba.$$

For single vector $\tilde{a} = a$.

Introduction to geometric algebra

Trigonometric expression for outer product

Outer product in coordinate form:

$$a \wedge b = (a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2) \wedge (b_1 \mathbf{e}_1 + b_2 \mathbf{e}_2),$$

$$a \wedge b = a_1 b_1 (\mathbf{e}_1 \wedge \mathbf{e}_1) + a_1 b_2 (\mathbf{e}_1 \wedge \mathbf{e}_2) + a_2 b_1 (\mathbf{e}_2 \wedge \mathbf{e}_1) + a_2 b_2 (\mathbf{e}_2 \wedge \mathbf{e}_2).$$

Because $\mathbf{e}_1 \wedge \mathbf{e}_1 = \mathbf{e}_2 \wedge \mathbf{e}_2 = 0$ and $\mathbf{e}_1 \wedge \mathbf{e}_2 = -\mathbf{e}_2 \wedge \mathbf{e}_1$, we have

$$a \wedge b = (a_1 b_2 - a_2 b_1) \mathbf{e}_1 \wedge \mathbf{e}_2 = \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{e}_1 \wedge \mathbf{e}_2.$$

Scalar $a_1 b_2 - a_2 b_1 = S$ is the area $S = |a||b| \sin \varphi$ of parallelepiped made of vectors a and b . Then it is clear that

$$a \wedge b = S \mathbf{e}_{12} = |a||b| \sin \varphi \mathbf{e}_{12}.$$

Introduction to geometric algebra

Trigonometric expression for outer product (other way)

Using $ab = a \cdot b + a \wedge b$ ir $ba = a \cdot b - a \wedge b$, we can write

$$abba = (a \cdot b)^2 - (a \wedge b)^2.$$

$bb = \tilde{b}b = |b|^2$ is the scalar, which commutes with a . This gives an expression for $(a \wedge b)^2$

$$(a \wedge b)^2 = -|a|^2|b|^2(1 - \cos^2 \varphi) = -|a|^2|b|^2 \sin^2 \varphi.$$

Because $e_{12}^2 = -1$ we have the same result.

Introduction to geometric algebra

Exponent of bivector

From inner and outer expressions we can write

$$ab = a \cdot b + a \wedge b = |a||b|(\cos \varphi + \mathbf{e}_{12} \sin \varphi).$$

Define the exponential

$$\cos \varphi + \mathbf{e}_{12} \sin \varphi = e^{\mathbf{e}_{12} \varphi}.$$

Check: expand in series and use the property $\mathbf{e}_{12}^2 = -1$ to write the exponential a sum of even and odd terms of $\mathbf{e}_{12}$ powers

$$e^{\mathbf{e}_{12} \varphi} = \sum_{n=0}^{\infty} \frac{(\mathbf{e}_{12} \varphi)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n \varphi^{2n}}{(2n)!} + \mathbf{e}_{12} \sum_{n=0}^{\infty} \frac{(-1)^{n+1} \varphi^{2n+1}}{(2n+1)!}.$$

Even and odd series give $\cos \varphi$ and $\sin \varphi$. So, $ab = |a||b|e^{\mathbf{e}_{12} \varphi}$.
This reminds the complex numbers $e^{i\varphi} = \cos \varphi + i \sin \varphi$.

Introduction to geometric algebra

Mathematica example

Example: derivation formula for $\sin(\alpha + \beta)$

For product of exponentials we can write (the same $\mathbf{e}_{12}$ in the exponent)

$$e^{\mathbf{e}_{12}\alpha} e^{\mathbf{e}_{12}\beta} = e^{\mathbf{e}_{12}(\alpha+\beta)}$$

$$\begin{aligned} e^{\mathbf{e}_{12}\alpha} e^{\mathbf{e}_{12}\beta} &= (\cos \alpha + \mathbf{e}_{12} \sin \alpha)(\cos \beta + \mathbf{e}_{12} \sin \beta) \\ &= \cos \alpha \cos \beta - \sin \alpha \sin \beta + \mathbf{e}_{12}(\sin \alpha \cos \beta + \sin \beta \cos \alpha) \end{aligned}$$

$$e^{\mathbf{e}_{12}(\alpha+\beta)} = \cos(\alpha + \beta) + \mathbf{e}_{12} \sin(\alpha + \beta)$$

Comparing terms at scalar and bivector $\mathbf{e}_{12}$ we see that

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta \text{ and}$$

$$\sin(\alpha + \beta) = (\sin \alpha \cos \beta + \sin \beta \cos \alpha).$$

Introduction to geometric algebra

Rotor

When a and b are of unit length ($|a| = |b| = 1$), their geometric product is exponent of bivector

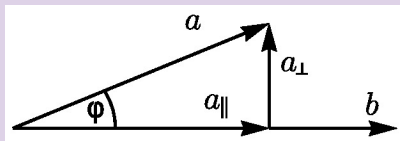
$$\hat{a}\hat{b} \equiv R(\varphi) = e^{\mathbf{e}_{12}\varphi}, \quad \mathbf{e}_{12}^2 = -1. \quad Cl_{2,0} \text{ ROTOR}$$

This is very similar to complex exponents $e^{i\varphi}$, therefore it plays an important role in rotations. Rotor is a **scalar+bivector**

Rotation **axes** is a misleading concept. Only planes of rotation, i. e. planes where rotation takes place, are important. Example: 2D or 4D (Lorentz transform is a rotation in $Cl_{1,3}$).

Introduction to geometric algebra

Orthogonal projection



Vector a orthogonal projection $a_{\parallel}$ and rejection $a_{\perp}$ with respect to vector b

Vector a projection $a_{\parallel}$ onto vector b . Unit vector is $\hat{b} = b/|b|$. Norm of a projection $|a_{\parallel}| = |a| \cos \varphi$. Therefore

$$a_{\parallel} = |a_{\parallel}| \hat{b} = |a||b| (b/|b|^2) \cos \varphi.$$

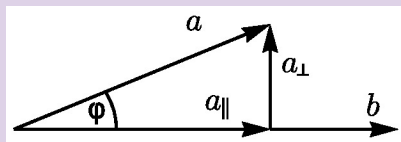
Using $b/|b|^2 = b^{-1}$ and $|a||b| \cos \varphi = a \cdot b$ yields

$$a_{\parallel} = (a \cdot b) b^{-1}. \quad \text{PROJECTION}$$

Note, that projection does not contain vector b length. Therefore, $a_{\parallel} = (a \cdot \hat{b}) \hat{b}^{-1} = (a \cdot \hat{b}) \hat{b}$ that can be generalized to project onto any multivector.

Introduction to geometric algebra

Orthogonal rejection



Vector $a_{\perp}$ is called rejection.

$$a_{\perp} = a - a_{\parallel} = a - (a \cdot b)b^{-1} = (ab - a \cdot b)b^{-1}.$$

Term in brackets is outer product, therefore

$$a_{\perp} = (a \wedge b)b^{-1}. \quad \text{REJECTION}$$

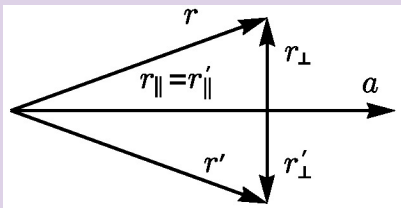
Summing projection and rejection yields initial vector a

$$a_{\parallel} + a_{\perp} = (a \cdot b)b^{-1} + (a \wedge b)b^{-1} = (a \cdot b + a \wedge b)b^{-1} = abb^{-1} = a.$$

Projection and rejection are **vectors**, while in traditional vector algebra they are understood as scalars. Projection and rejection vectors are orthogonal $a_{\parallel} \cdot a_{\perp} = \frac{1}{2}(a_{\parallel}a_{\perp} + a_{\perp}a_{\parallel}) = 0$.

Introduction to geometric algebra

Reflection



Reflection of r with respect to vector a

Parallel component remains unchanged, while that orthogonal to a change sign

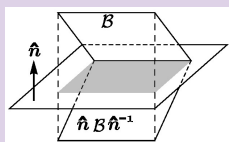
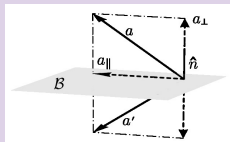
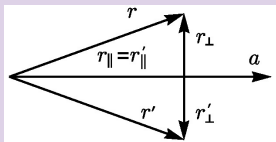
$$r' = r_{\parallel}' + r_{\perp}' = r_{\parallel} - r_{\perp}.$$

Inserting expression for projection $r_{\parallel}$ and rejection $r_{\perp}$ we have

$$\begin{aligned} r' &= (r \cdot a)a^{-1} - (r \wedge a)a^{-1} \\ &= (r \cdot a - r \wedge a)a^{-1} = (a \cdot r + a \wedge r)a^{-1} \\ &= ara^{-1}. \end{aligned}$$

Introduction to geometric algebra

Reflection (continued)



Actually important is only vector a direction because $a = |a|\hat{a}$ and $a^{-1} = \hat{a}/|a|$. Therefore the scalar $|a|$ disappear. So

$$r' = \hat{a} r \hat{a}. \quad \text{REFLECTION}$$

This is one of most beautiful formula in GA. For bivector reflection in 3D (**note the minus sign**)

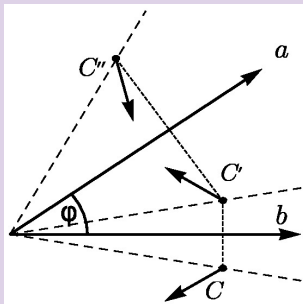
$$B' = (-\hat{n} a \hat{n}) \wedge (-\hat{n} b \hat{n}).$$

and after simplification

$$B' = \frac{1}{2}(\hat{n} a \hat{n} \hat{n} b \hat{n} - \hat{n} b \hat{n} \hat{n} a \hat{n}) = \frac{1}{2}\hat{n}(ab - ba)\hat{n} = \hat{n} B \hat{n}.$$

Introduction to geometric algebra

Combination of $2n$ reflections = rotation (Cartan)

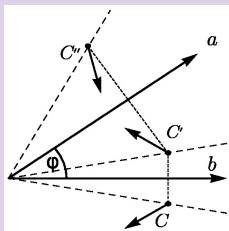


Two reflections, one with respect to b , other with respect to a . $C \rightarrow C' \rightarrow C''$. First reflection: $r \rightarrow r' = \hat{b}r\hat{b}^{-1}$. Second reflection $r' \rightarrow r'' = \hat{a}r'\hat{a}^{-1}$. Composition results in

$$\begin{aligned} r'' &= \hat{a}(\hat{b}r\hat{b}^{-1})\hat{a}^{-1} = (\hat{a}\hat{b})r(\hat{b}^{-1}\hat{a}^{-1}) \\ &= (\hat{a}\hat{b})r(\hat{a}\hat{b})^{-1}. \end{aligned}$$

Introduction to geometric algebra

Rotation



Rewrite using formula for exponent

$$r'' = e^{\mathbf{e}_{12}\varphi} r (e^{\mathbf{e}_{12}\varphi})^{-1}.$$

$\exp(-\mathbf{e}_{12}\varphi) \exp(\mathbf{e}_{12}\varphi) = (\cos \varphi - \mathbf{e}_{12} \sin \varphi)(\cos \varphi + \mathbf{e}_{12} \sin \varphi) = 1$,
then $(e^{\mathbf{e}_{12}\varphi})^{-1} = e^{-\mathbf{e}_{12}\varphi}$ and can rewrite as

$$r'' = e^{\mathbf{e}_{12}\varphi} r e^{-\mathbf{e}_{12}\varphi}.$$

The last formula works in **any** dimension!

1 Introduction to geometric algebra

2 $Cl_{3,0}$ quantum mechanics

- Standard quantum mechanics
- How to dispense with matrices?
- Two level model
- Example of monolayer graphene
- The algorithm

3 Multivector inverse for $n = p + q \leq 6$

- Introduction
- Grade-negated self-product
- Inverse of general MV of algebras of dimension $n = p + q \leq 5$
- Inverse of general MV of algebras of dimension $n = p + q = 6$

4 Coordinate-free solution of quaternionic Sylvester equation

5 Conclusions

Standard quantum mechanics approach

Two level model

The Hamiltonian

$$\hat{H} = \begin{bmatrix} \varepsilon_0 - \varepsilon_3 & \varepsilon_1 - i\varepsilon_2 \\ \varepsilon_1 + i\varepsilon_2 & \varepsilon_0 + \varepsilon_3 \end{bmatrix},$$

where ε_i are reals. Pauli matrices

$$\hat{\sigma}_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \hat{\sigma}_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \hat{\sigma}_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},$$

are $Cl_{3,0}$ vector representations, because

$$\hat{\sigma}_i \hat{\sigma}_j + \hat{\sigma}_j \hat{\sigma}_i = \begin{cases} 0, & \text{if } i \neq j \\ 2\hat{1}, & \text{if } i = j \end{cases}, \quad \text{where } \hat{1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

and $\hat{\sigma}_1^2 = \hat{\sigma}_2^2 = \hat{\sigma}_3^2 = \hat{1}$. The solution of eigenequation is trivial.

Standard quantum mechanics approach

Electron in a monolayer graphene (Nobel 2010, A. Geim and K. Novoselov)

Graphene is 2D crystal with honeycomb lattice. In the Hilbert space formalism, the **massless Dirac Hamiltonian** in the vicinity of critical K point of graphene reduces to simple form

$$\hat{H}_{MG} = \hbar v_F \begin{bmatrix} 0 & k_x - i k_y \\ k_x + i k_y & 0 \end{bmatrix},$$

where $v_F \approx 10^6$ m/s is the Fermi velocity. The wave vectors $\mathbf{k} = (k_x, k_y)$ lies in the graphene plane and is measured with respect to energy valley minimum. In the following the natural units $\hbar = 1$ along with $v_F = 1$ are used.

Standard quantum mechanics approach

Eigenenergies of monolayer graphene

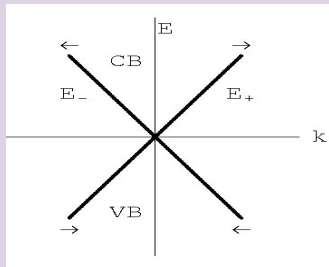
The eigenenergies of the Hamiltonian (27) are

$$E_{\pm} = \pm \sqrt{k_x^2 + k_y^2} = \pm |\mathbf{k}|,$$

where plus/minus sign corresponds to conduction/valence bands. Since $E \propto |\mathbf{k}|$, the spectrum is called ultrarelativistic.

Mathematics of pseudospin is similar to electron spin. Physically, it is a superposition of dangling p -orbitals of carbon atoms.

Spectrum of monolayer graphene



CB/VB means conduction/valence band. Left/right arrows show pseudospin direction.

$Cl_{3,0}$ algebra approach

Projection into base matrices

Calculate products of two matrices $\hat{\sigma}_1^2 = \hat{\sigma}_2^2 = \hat{\sigma}_3^2 = \hat{1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
and

$$\hat{\sigma}_1\hat{\sigma}_2 = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}, \quad \hat{\sigma}_2\hat{\sigma}_3 = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}, \quad \hat{\sigma}_3\hat{\sigma}_1 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

When trace is nonzero? From the Hamiltonian

$$\hat{H} = \begin{bmatrix} \varepsilon_0 + \varepsilon_3 & \varepsilon_1 - i\varepsilon_2 \\ \varepsilon_1 + i\varepsilon_2 & \varepsilon_0 - \varepsilon_3 \end{bmatrix},$$

ε_i can be extracted using

$$\varepsilon_i = \frac{1}{d} \text{Tr}(\hat{H}\hat{\sigma}_i), \quad d = 2.$$

$Cl_{3,0}$ algebra approach

Algebraic structure

Mathematica example

The result is

$$\hat{H} = \varepsilon_0 + \varepsilon_1 \hat{\sigma}_1 + \varepsilon_2 \hat{\sigma}_2 + \varepsilon_3 \hat{\sigma}_3 .$$

Any 2×2 hermitian matrix can be expanded in base of

$$\{1, \sigma, I\sigma, I\} = \{\text{scalar, 3 vectors, 3 bivectors, trivector (pseudoscalar)}\}$$

When formulating problem purely in geometric algebra remove hats from $\hat{\sigma}_i$, i.e.

$$H = \varepsilon_0 + \varepsilon_1 \sigma_1 + \varepsilon_2 \sigma_2 + \varepsilon_3 \sigma_3 .$$

Problem. $\hat{H}\psi = E\psi$. What to do with ψ ?

$Cl_{3,0}$ algebra approach

What is the spinor in GA?

Simplified idea: replace spinor by matrix

$$\begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix} \rightarrow \begin{bmatrix} \psi_1 & 0 \\ \psi_2 & 0 \end{bmatrix}$$

An ideal is a subset I of elements in a ring R that forms an additive group and has the property that, whenever x belongs to R and y belongs to I , then xy and yx belong to I . For example, the set of even integers is an ideal in the ring of integers $\mathbb{Z}$. Given an ideal I , it is possible to define a quotient ring R/I .

Analogy based on a low dimension $Cl_{p,q}$ Clifford algebras: spinor is a sum of even elements, i. e. scalar+bivector+ $\dots$
(not general enough, not quite true).

$Cl_{3,0}$ algebra approach

What is the spinor in GA? (continued)

Mathematica example (continued)

Replacement rules by even multivectors

$$\begin{bmatrix} a_0 + ia_3 \\ -a_2 + ia_1 \end{bmatrix} = |\psi\rangle \quad \Leftrightarrow \quad \psi = a_0 + a_1 I\sigma_1 + a_2 I\sigma_2 + a_3 I\sigma_3,$$

$Cl_{3,0}$ SPINOR

where a_i are any real numbers, which satisfy

$$\langle\psi|\psi\rangle = \tilde{\psi}\psi = \psi\tilde{\psi} = a_0^2 + a_1^2 + a_2^2 + a_3^2 = 1.$$

$\tilde{\psi}$ can be replaced by

$$[a_0 - ia_3, -a_2 - ia_1] = \langle\psi| \quad \Leftrightarrow \quad \tilde{\psi}.$$

However for different spinors $\langle\psi|\phi\rangle_0 \neq \tilde{\psi}\phi$.

$Cl_{3,0}$ algebra approach

Replacement rules

More replacement rules

Hilbert space	$Cl_{3,0}$ algebra	Hilbert space	$Cl_{3,0}$ algebra
$\hat{1} \psi\rangle$	ψ	$\hat{\sigma}_k \psi\rangle$	$\sigma_k \psi \sigma_3$
$i \psi\rangle$	$\psi I\sigma_3$	$i\hat{\sigma}_k \psi\rangle$	$I\sigma_k \psi$
$(a_r + ia_i) \psi\rangle$	$a_r \psi + a_i \psi I\sigma_3$	$\partial \psi\rangle/\partial t$	$\dot{\psi}$
$ \psi^*\rangle$	$\sigma_2 \psi \sigma_2$	$i\partial \psi\rangle/\partial t$	$I\dot{\psi} \sigma_3$
$\langle\psi $	$\tilde{\psi}$	$\hat{k}_x \psi\rangle \equiv -i \frac{\partial \psi\rangle}{\partial x}$	$-\frac{\partial\psi}{\partial x} I\sigma_3$

Relation between traditional Hilbert space formalism and $Cl_{3,0}$
geometric algebra

$Cl_{3,0}$ algebra approach

Two level model: reformulation

Replacement rule application: $\hat{H}|\psi\rangle$ is replaced by

$$H(\psi) = \varepsilon_0\psi + (\varepsilon_1\sigma_1 + \varepsilon_2\sigma_2 + \varepsilon_3\sigma_3)\psi\sigma_3.$$

Note the appearance of privileged direction σ_3 . Introduce the vector $\varepsilon = \varepsilon_1\sigma_1 + \varepsilon_2\sigma_2 + \varepsilon_3\sigma_3$,

$$H(\psi) = \varepsilon_0\psi + \varepsilon\psi\sigma_3.$$

Solve GA eigenequation with noncommuting quantities

$$H(\psi) = \varepsilon\psi \quad \text{or} \quad \varepsilon_0\psi_{\pm} + \varepsilon\psi_{\pm}\sigma_3 = \varepsilon_{\pm}\psi_{\pm}.$$

Rewrite it as an rotor equation, i. e. multiply from right by σ_3 and then by $\tilde{\psi}_{\pm}$:

$$\varepsilon_0\psi_{\pm}\sigma_3\tilde{\psi}_{\pm} + \varepsilon\psi_{\pm}\tilde{\psi}_{\pm} = \varepsilon_{\pm}\psi_{\pm}\sigma_3\tilde{\psi}_{\pm}.$$

$Cl_{3,0}$ algebra approach

Two level model: eigenvalues

Look for spinor satisfying $\psi_{\pm} \tilde{\psi}_{\pm} = 1$. Then above formula simplifies to an elegant GA form

$$\psi_{\pm} \sigma_3 \tilde{\psi}_{\pm} = \frac{\epsilon}{\epsilon_{\pm} - \epsilon_0}.$$

Due to normalization requirement left side is an rotor, which rotates vector σ_3 into vector ϵ . Length of σ_3 is unity and cannot change under rotation. Therefore square of right hand side is

$$\left(\frac{\epsilon}{\epsilon_{\pm} - \epsilon_0} \right)^2 = \frac{\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2}{(\epsilon_{\pm} - \epsilon_0)^2} = 1.$$

This is the **quantization condition**! After solving it with respect to $\epsilon_{\pm}$ we obtain eigenvalues

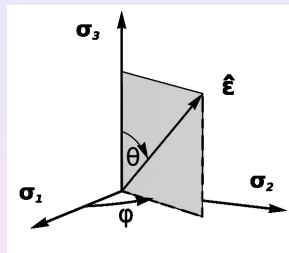
$$\epsilon_{\pm} = \epsilon_0 \pm |\epsilon| = \epsilon_0 \pm \sqrt{\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2}.$$

$Cl_{3,0}$ algebra approach

Two level model: eigenvectors

After eigenvalues have been found, one can solve for eigenvectors ψ_+ and ψ_- ,

$$\psi_{\pm} \sigma_3 \tilde{\psi}_{\pm} = \pm \frac{\epsilon}{|\epsilon|} \equiv \pm \hat{\epsilon}.$$



Because initial σ_3 and final $\hat{\epsilon}$ unit vectors are known, the spinors-rotors can be easily constructed

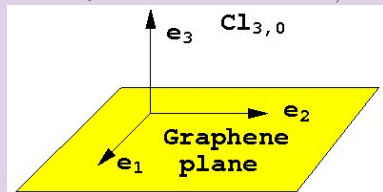
$$\psi_+ = \exp\left(-\frac{\sigma_3 \wedge \hat{\epsilon}}{|\sigma_3 \wedge \hat{\epsilon}|} \frac{\theta}{2}\right) = \cos \frac{\theta}{2} - \frac{\sigma_3 \wedge \hat{\epsilon}}{|\sigma_3 \wedge \hat{\epsilon}|} \sin \frac{\theta}{2},$$

Here θ denotes angle between vectors σ_3 and $\hat{\epsilon}$. Rotation angle is from σ_3 to $\hat{\epsilon}$. $\cos \varphi = \epsilon_1/(\epsilon_1^2 + \epsilon_2^2)^{1/2}$.

$Cl_{3,0}$ algebra approach

$Cl_{3,0}$ algebra applied to graphene

Minimal algebra to describe the monolayer Hamiltonian is $Cl_{3,0}$.



Base vectors $\mathbf{e}_1^2 = \mathbf{e}_2^2 = \mathbf{e}_3^2 = 1$ play the role of space coordinates. The orthogonal oriented planes are bivectors $\mathbf{Ie}_1 = \mathbf{e}_2\mathbf{e}_3$, $\mathbf{Ie}_2 = \mathbf{e}_3\mathbf{e}_1$, and $\mathbf{Ie}_3 = \mathbf{e}_1\mathbf{e}_2$. With scalar and pseudoscalar $I = \mathbf{e}_1\mathbf{e}_2\mathbf{e}_3$ included, there are 8 basis elements.

Replacement rules

Transform the Hamiltonian (27) to $Cl_{3,0}$ using mapping rules from Hilbert to Clifford space [table]

$$|\psi\rangle \Leftrightarrow \psi = a_0 + a_1\mathbf{Ie}_1 + a_2\mathbf{Ie}_2 + a_3\mathbf{Ie}_3,$$

$$\lambda|\psi\rangle \Leftrightarrow \lambda\psi,$$

$$\hat{\sigma}_i|\psi\rangle \Leftrightarrow \mathbf{e}_i\psi\mathbf{e}_3.$$

λ is the scalar and $\hat{\sigma}_i$ are Pauli matrices. Then (27), $\hat{H}_{MG}|\psi\rangle$, maps to the following simple GA function for spinor ψ

$$\mathbf{H}_{MG}(\psi) = \mathbf{k}\psi\mathbf{e}_3. \quad (1)$$

The wave vector $\mathbf{k} = k_x\mathbf{e}_1 + k_y\mathbf{e}_2$ lies in the graphene plane. $\mathbf{e}_3$ represents the pseudospin quantization axis, which is perpendicular to graphene plane.

$Cl_{3,0}$ algebra approach

Eigenvalue equation

From (1), eigenvalue equation for monolayer graphene follows

$$\mathbf{k}\psi\mathbf{e}_3 = \psi E,$$

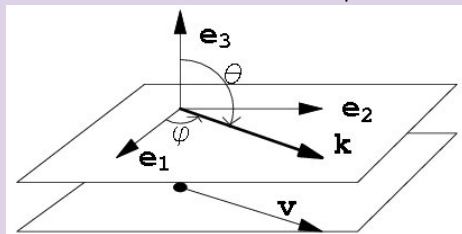
which can be easily rewritten as an equation for rotor

$$\frac{\psi\mathbf{e}_3\tilde{\psi}}{\psi\tilde{\psi}} = \frac{\mathbf{k}}{|\mathbf{k}|^2}E, \quad (2)$$

where $|\mathbf{k}| \equiv k = \sqrt{k_x^2 + k_y^2}$ is the magnitude of the wave vector. Since $\mathbf{e}_3$ is a unit vector, after rotation by ψ we shall obtain the unit vector on the right hand side too.

Eigenvalues

This means that the square of right hand side of (2) must be 1, $(E/k)^2 = 1$, from which follows 2D ultrarelativistic dispersion law $E_{\pm} = \pm k = \pm\sqrt{k_x^2 + k_y^2}$.



Electron with velocity $\mathbf{v}$ and coordinate system for graphene. The planes represent graphene layers. $\mathbf{k}$ is electron wave vector. $\theta = \pi/2$, $\cos \varphi = k_1/|\mathbf{k}|$, $\sin \varphi = k_2/|\mathbf{k}|$.

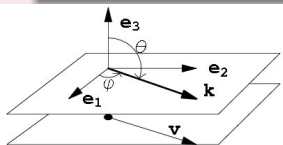
$Cl_{3,0}$ algebra approach

Eigenmultivectors of monolayer graphene

The corresponding eigenmultivectors can be deduced from the same equation (2) after insertion of eigenvalue $E_{\pm}$ back and requiring the **rotor-spinor** ψ to satisfy $\psi\tilde{\psi} = 1$. Then, the equation (2) reduces to

$$\psi_{\pm} \mathbf{e}_3 \tilde{\psi}_{\pm} = \pm \hat{\mathbf{k}}, \quad (3)$$

where $\hat{\mathbf{k}} = \mathbf{k}/|\mathbf{k}|$. Thus, according to the geometric algebra the eigenspinors $\psi_{\pm}$ of a monolayer graphene are rotors that bring the unit vector $\mathbf{e}_3$ that is parallel to quantization axis to new direction $\pm \hat{\mathbf{k}}$, which is either parallel or antiparallel to the electron wave vector $\mathbf{k}$.



No such clear-cut geometrical interpretation exists for eigenspinors calculated in the Hilbert space.

$Cl_{3,0}$ algebra approach

Eigenmultivectors of monolayer graphene

The solution of (3) can be written automatically as a rotor in the unit bivector $\hat{\mathbf{k}} \wedge \mathbf{e}_3$ plane,

$$\psi_{\pm} = e^{\pm \hat{\mathbf{k}} \wedge \mathbf{e}_3 \pi/4} = \frac{1}{\sqrt{2}} (1 \pm \hat{\mathbf{k}} \wedge \mathbf{e}_3). \quad (4)$$

Since $\hat{\mathbf{k}}$ and $\mathbf{e}_3$ are mutually perpendicular the outer product may be replaced by geometric one, $\hat{\mathbf{k}} \wedge \mathbf{e}_3 = \hat{\mathbf{k}} \mathbf{e}_3$. In the expanded coordinate form the eigenspinors are

$$\psi_{\pm} = \frac{1}{\sqrt{2}} (1 \pm \sin \varphi I \mathbf{e}_1 \mp \cos \varphi I \mathbf{e}_2). \quad (5)$$

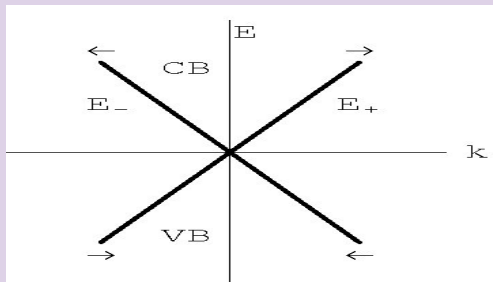
The eigenspinors (4) are written in a coordinate-free way. The appearance of $\mathbf{e}_3$ in (4) is indispensable since the quantum mechanics requires to indicate the direction of the pseudospin (or spin) quantization axis which in our case coincides with $\mathbf{e}_3$.

$Cl_{3,0}$ algebra approach

Eigenmultivectors of monolayer graphene

The spin is an intrinsic property of electron, while the pseudospin is the property of interacting p -orbitals of carbon atoms that make up the lattice. The average pseudospin vector may be found in exactly the same way as the electron spin,

$$\langle \mathbf{P} \rangle = \psi_{\pm} \mathbf{e}_3 \tilde{\psi}_{\pm} = \pm \mathbf{k} / |\mathbf{k}| = \pm (\cos \varphi \mathbf{e}_1 + \sin \varphi \mathbf{e}_2). \quad (6)$$



Graphene spectrum (eigenstates). The pseudospin is either parallel or antiparallel to electron propagation direction.

The algorithm

Main steps

- Using replacement rules rewrite the hamiltonian in terms of geometric algebra
- Put obtained equation into rotor equation form
- Solve algebraic equation for eigenenergy
- Substitute eigenenergy into rotor equation and solve for eigenmultivector.

The algorithm works for Hamiltonians of electron in a quantum well ($Cl_{3,0}$ algebra) as well as for similar problems formulated in $Cl_{1,3}$ and $Cl_{3,1}$ algebras. In particular, we have applied the method to a pair of coupled two-level atoms and a bilayer graphene.

The algorithm

Main steps

- Using replacement rules rewrite the hamiltonian in terms of geometric algebra
- Put obtained equation into rotor equation form
- Solve algebraic equation for eigenenergy
- Substitute eigenenergy into rotor equation and solve for eigenmultivector.

The algorithm works for Hamiltonians of electron in a quantum well ($Cl_{3,0}$ algebra) as well as for similar problems formulated in $Cl_{1,3}$ and $Cl_{3,1}$ algebras. In particular, we have applied the method to a pair of coupled two-level atoms and a bilayer graphene.

The algorithm

Main steps

- Using replacement rules rewrite the hamiltonian in terms of geometric algebra
- Put obtained equation into rotor equation form
- Solve algebraic equation for eigenenergy
- Substitute eigenenergy into rotor equation and solve for eigenmultivector.

The algorithm works for Hamiltonians of electron in a quantum well ($Cl_{3,0}$ algebra) as well as for similar problems formulated in $Cl_{1,3}$ and $Cl_{3,1}$ algebras. In particular, we have applied the method to a pair of coupled two-level atoms and a bilayer graphene.

The algorithm

Main steps

- Using replacement rules rewrite the hamiltonian in terms of geometric algebra
- Put obtained equation into rotor equation form
- Solve algebraic equation for eigenenergy
- Substitute eigenenergy into rotor equation and solve for eigenmultivector.

The algorithm works for Hamiltonians of electron in a quantum well ($Cl_{3,0}$ algebra) as well as for similar problems formulated in $Cl_{1,3}$ and $Cl_{3,1}$ algebras. In particular, we have applied the method to a pair of coupled two-level atoms and a bilayer graphene.

1 Introduction to geometric algebra

2 $Cl_{3,0}$ quantum mechanics

- Standard quantum mechanics
- How to dispense with matrices?
- Two level model
- Example of monolayer graphene
- The algorithm

3 Multivector inverse for $n = p + q \leq 6$

- Introduction
- Grade-negated self-product
- Inverse of general MV of algebras of dimension $n = p + q \leq 5$
- Inverse of general MV of algebras of dimension $n = p + q = 6$

4 Coordinate-free solution of quaternionic Sylvester equation

5 Conclusions

Problem formulation

- General Clifford algebra multivector (MV)
 $A = \text{scalar} + \text{bivector} + \dots + \text{pseudoscalar}$ is considered
- The problem is to find the inverse of A , i. e. A^{-1} for which $AA^{-1} = A^{-1}A = 1$ holds,
- $n = p + q$ in the title indicates that algorithm for A^{-1} depends only on dimension n of the Clifford algebra $Cl_{p,q}$ and not on a particular signature (p, q)
- The search of required algorithm heavily relies on computer-aided symbolic mathematics (*Mathematica*, *Maple*, *Reduce*)

So, before explaining calculation of inverse of MV a brief outlook of the *Mathematica* package that was written for this purpose will be presented.

GeometricAlgebra package for *Mathematica*

Main features

- Textbook notation (implemented without *Mathematica* palettes) and automatic precedence of inner→outer→geometric products
- Algebraic operations in orthonormal frame (additive representation, symbolic coefficients)
- Switching between multiple algebras (stained in different colors) during *Mathematica* session
- Possibility for matrix representations of $Cl_{p,q}$ algebra multivectors (two methods)
- Idempotents with basis element sorting (Lex, RevLex etc.)
- Main involutions and general multivector inverse
- Ideal spinors (left ideals, ...)

Files

- GAC_Base.nb (main file with installation information)
- UnicodeCharactersAdd.tr (add-on for *Mathematica* system file)
- A new version is planned in spring.

May be download under licence GPL-3.0 from

<https://github.com/ArturasAcus/GeometricAlgebra>

Grade-negated self-product

Grade- r negation operation

- Changes the sign of grade- r part of the multivector A
- It is an **involution** denoted as $A_{\bar{r}}$. Note the bar over negated grade index.
- Formal definition

$$A_{\bar{r}} = A - 2\langle A \rangle_r$$
- Main properties:

$$(A_{\bar{p}})_{\bar{r}} = A_{\bar{p},\bar{r}} = A_{\bar{r},\bar{p}}, \quad A_{\bar{r},\bar{r}} = A,$$

$$(A + B)_{\bar{r}} = A_{\bar{r}} + B_{\bar{r}}$$
- However, $(AB)_{\bar{r}} \neq A_{\bar{r}}B_{\bar{r}}$
- Commutator

$$AA_{\bar{r}} - A_{\bar{r}}A = 2(\langle A \rangle_r A - A \langle A \rangle_r)$$

Mathematica example

Relation to standard involutions

- Reversion $\tilde{A} = A_{\bar{2},\bar{3},\bar{6},\bar{7},\bar{10},\bar{11}} \dots$
- Grade inversion

$$\hat{A} = A_{\bar{1},\bar{3},\bar{5},\bar{7},\bar{9},\bar{11}} \dots$$
- Clifford conjugate

$$\tilde{\tilde{A}} = A_{\bar{1},\bar{2},\bar{5},\bar{6},\bar{9},\bar{10}} \dots$$

Important: grades 0; 4; 8; 12; $\dots$ are absent in the above list of standard GA involutions. They play special role in inverse formulas. Example:

$$A = 1 + 2\mathbf{e}_1 + 3\mathbf{e}_2 + \mathbf{e}_{23}$$

$$A_{\bar{1}} = 1 - 2\mathbf{e}_1 - 3\mathbf{e}_2 + \mathbf{e}_{23}$$

$$A_{\bar{2}} = 1 + 2\mathbf{e}_1 + 3\mathbf{e}_2 - \mathbf{e}_{23}$$

Grade-negated self-product

Definitions and the idea

Grade-negated self-product is defined via grade- r negation operation.

- Grade-negated self-product of general multivector A is geometric product of A with any number of its grade-negated counterparts. Examples $A A_{\bar{r}} A_{\bar{k}, \bar{l}, \bar{t}} A_{\bar{s}, \bar{t}} \cdots$ or $A_{\bar{r}} A_{\bar{k}, \bar{l}, \bar{t}} A_{\bar{s}, \bar{t}} \cdots A$.
- Self-products where the initial MV A stands in the left-most or in the right-most position are of special importance in finding the inverse multivector A^{-1} .
- Any expression which consists only of operations of geometric product and grade negation will be true in any other $p + q$ dimensional algebra $Cl_{p,q}$ independent of signature (p, q) . Thus, the algorithm depends on vector space dimension n only.
- Assume we can construct a **scalar** s_m using just grade-negated self-products of m multivectors with the initial multivector A standing either in the left-most or right-most position.

Grade-negated self-product

Definitions and the idea (continued)

- In this way obtained scalar call the **determinant norm** of A.

scalar : $s_m = c_1 \overbrace{AA_{\bar{r},\bar{t},\dots}A_{\bar{s},\dots}}^m \cdots + c_2 \overbrace{AA_{\bar{g},\bar{j},\dots}A_{\bar{p},\dots}}^m \cdots + \cdots$

- If we succeed in constructing such a scalar then left/right inverse of MV can be obtained by removing left or right most positioned initial MV A from the scalar

$s_m = c_1 AA_{\bar{r},\bar{t},\dots}A_{\bar{s},\dots} \cdots + c_2 AA_{\bar{g},\bar{j},\dots}A_{\bar{p},\dots} \cdots + \cdots$ and then dividing the result by s_m :

$$A^{-1} = \sum \frac{A_{\bar{r},\bar{t},\dots}A_{\bar{s},\dots} \cdots}{s_m}.$$

Inverses in dimension $n = 2$

Two dimensional algebras: $Cl_{2,0}$ case

- Write $A = \sum_{J=0}^{2^n-1} a_J \mathbf{e}_J$, where the multi-index J covers all orthonormal base elements arranged in inverse degree lexicographic order, i. e., the lowest grade elements appear first while elements of the same grade are ordered lexicographically:
 $J = [\{\}, \{1\}, \{2\}, \{1, 2\}]$.
- Check that self-product $AA_{\bar{1},\bar{2}}$ immediately gives the required scalar
 $s_2 = a_{\{\}}^2 - a_1^2 - a_2^2 + a_{1,2}^2$.

Coefficient notation

- It is more convenient to rewrite the coefficient indices as $s_2 = a_0^2 - a_1^2 - a_2^2 + a_3^2$, i. e. the index of scalar a_0 is zero, the coefficients at vectors run from 1 to n while numerical indices of higher grades increase monotonically up to 2^n .
- General rule for grade- r index: start the element index by $\sum_{k=0}^{r-1} \binom{n}{k}$ and end by $\left(\sum_{k=0}^{r-1} \binom{n}{k}\right) + \binom{n}{r} - 1$. Here $\binom{n}{k}$ is the binomial coefficient.

Inverses in dimension $n = 2$

Notes about determinant norm

- The determinant norm s_m comprises all coefficients a_i of multivector A
- The MV has inverse if determinant norm is unequal to zero, $s_m \neq 0$. This imposes the requirement for MV coefficients.
- Because reverse leaves the determinant norm invariant, the formula for $n = 2$ can be written as $s_2 = AA_{\bar{1},\bar{2}}$ and $s'_2 = \widetilde{\tilde{A}}_{\bar{1},\bar{2}}\tilde{A}$, from which the left/right inverse formulas follow,

$$A^{-1} = A_{\bar{1},\bar{2}}/(AA_{\bar{1},\bar{2}}); \quad \widetilde{A}^{-1} = \widetilde{\tilde{A}}_{\bar{1},\bar{2}}/(\widetilde{\tilde{A}}_{\bar{1},\bar{2}}\tilde{A}).$$

- In all cases the highest degree of coefficients in the determinant norm always matches the degree of determinant of real matrix representation of A , in this case 2×2 real matrices (next slide)

Inverses in dimension $n = 2$

Notes about determinant norm

- The determinant norm s_m comprises all coefficients a_i of multivector A
- The MV has inverse if determinant norm is unequal to zero, $s_m \neq 0$. This imposes the requirement for MV coefficients.
- Because reverse leaves the determinant norm invariant, the formula for $n = 2$ can be written as $s_2 = AA_{\bar{1},\bar{2}}$ and $s'_2 = \widetilde{A}_{\bar{1},\bar{2}}\widetilde{A}$, from which the left/right inverse formulas follow,

$$A^{-1} = A_{\bar{1},\bar{2}}/(AA_{\bar{1},\bar{2}}); \quad \widetilde{A}^{-1} = \widetilde{A}_{\bar{1},\bar{2}}/(\widetilde{A}_{\bar{1},\bar{2}}\widetilde{A}).$$

- In all cases the highest degree of coefficients in the determinant norm always matches the degree of determinant of real matrix representation of A , in this case 2×2 real matrices (next slide)

Inverses in dimension $n = 2$

Notes about determinant norm

- The determinant norm s_m comprises all coefficients a_i of multivector A
- The MV has inverse if determinant norm is unequal to zero, $s_m \neq 0$. This imposes the requirement for MV coefficients.
- Because reverse leaves the determinant norm invariant, the formula for $n = 2$ can be written as $s_2 = AA_{\bar{1},\bar{2}}$ and $s'_2 = \widetilde{\tilde{A}}_{\bar{1},\bar{2}}\tilde{A}$, from which the left/right inverse formulas follow,

$$A^{-1} = A_{\bar{1},\bar{2}}/(AA_{\bar{1},\bar{2}}); \quad \widetilde{A}^{-1} = \tilde{A}_{\bar{1},\bar{2}}/(\tilde{A}_{\bar{1},\bar{2}}\tilde{A}).$$

- In all cases the highest degree of coefficients in the determinant norm always matches the degree of determinant of real matrix representation of A , in this case 2×2 real matrices (next slide)

Inverses in dimension $n = 2$

Notes about determinant norm

- The determinant norm s_m comprises all coefficients a_i of multivector A
- The MV has inverse if determinant norm is unequal to zero, $s_m \neq 0$. This imposes the requirement for MV coefficients.
- Because reverse leaves the determinant norm invariant, the formula for $n = 2$ can be written as $s_2 = AA_{\bar{1},\bar{2}}$ and $s'_2 = \widetilde{A}_{\bar{1},\bar{2}}\widetilde{A}$, from which the left/right inverse formulas follow,

$$A^{-1} = A_{\bar{1},\bar{2}}/(AA_{\bar{1},\bar{2}}); \quad \widetilde{A}^{-1} = \widetilde{A}_{\bar{1},\bar{2}}/(\widetilde{A}_{\bar{1},\bar{2}}\widetilde{A}).$$

- In all cases the highest degree of coefficients in the determinant norm always matches the degree of determinant of real matrix representation of A , in this case 2×2 real matrices (**next slide**)

Inverses in dimension $n = 2$

Period-8 table of irreducible matrix representations of GA for $Cl_{p,q}$, where p and q indicate positive and negative signature.

$p \backslash q$	0	1	2	3	4	5	6	7
0	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{H}$	${}^2\mathbb{H}$	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$
1	${}^2\mathbb{R}$	$\mathbb{R}(2)$	$\mathbb{C}(2)$	$\mathbb{H}(2)$	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$
2	$\mathbb{R}(2)$	${}^2\mathbb{R}(2)$	$\mathbb{R}(4)$	$\mathbb{C}(4)$	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$
3	$\mathbb{C}(2)$	$\mathbb{R}(4)$	${}^2\mathbb{R}(4)$	$\mathbb{R}(8)$	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$
4	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$	$\mathbb{R}(16)$	$\mathbb{C}(16)$	$\mathbb{H}(16)$	${}^2\mathbb{H}(16)$
5	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$	${}^2\mathbb{R}(16)$	$\mathbb{R}(32)$	$\mathbb{C}(32)$	$\mathbb{H}(32)$
6	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$	$\mathbb{R}(32)$	${}^2\mathbb{R}(32)$	$\mathbb{R}(64)$	$\mathbb{C}(64)$
7	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$	$\mathbb{C}(32)$	$\mathbb{R}(64)$	${}^2\mathbb{R}(64)$	$\mathbb{R}(128)$

The left superscript denotes direct sum of matrices, ${}^2\mathbb{R}(2) \equiv \mathbb{R}(2) \oplus \mathbb{R}(2)$.

Inverse formulas for $n = p + q \leq 6$

$Cl_{p,q}$	A^{-1}	
$p + q = 0$	$\frac{B}{AB},$	$B = 1$
$p + q = 1$	$\frac{B(AB)_{\bar{1}}}{AB(AB)_{\bar{1}}} = \frac{\widehat{A}}{AA},$	$B = 1$
$p + q = 2$	$\frac{C}{AC} = \frac{\widetilde{\widehat{A}}}{AA},$	$C = A_{\bar{1},\bar{2}}$
$p + q = 3$	$\frac{C(AC)_{\bar{3}}}{AC(AC)_{\bar{3}}} = \frac{\widetilde{\widetilde{\widehat{AAA}}}}{AAAA},$	$C = A_{\bar{1},\bar{2}}$
$p + q = 4$	$\frac{D}{AD},$	$D = A_{\bar{2},\bar{3}}(AA_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}$ or $D = A_{\bar{1},\bar{2}}(AA_{\bar{1},\bar{2}})_{\bar{3},\bar{4}}$
$p + q = 5$	$\frac{D(AD)_{\bar{5}}}{AD(AD)_{\bar{5}}},$	$D = A_{\bar{2},\bar{3}}(AA_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}$
$p + q = 6$	$\frac{G}{AG}, \quad G = \frac{1}{3}A_{\bar{2},\bar{3},\bar{6}}\left(H(HH)_{\bar{1},\bar{4},\bar{5}} + 2(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}}\right)$ or $G = \frac{1}{3}A_{\bar{2},\bar{3},\bar{6}}\left((H(HH)_{\bar{1},\bar{5}})_{\bar{4}}\right)_{\bar{1},\bar{5}} + 2(H_{\bar{4},\bar{5}}(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{4}})_{\bar{1},\bar{4}}$ with $H = AA_{\bar{2},\bar{3},\bar{6}} = \widetilde{AA}$	

Summary of formulas for inverse multivectors A^{-1} in Clifford algebras of dimension $n \leq 6$.

Inverses in dimension $n = p + q = 5$

Inverse formulas are not unique for $n > 4$. Case $n = 5$.

$N \downarrow$	Abbreviation $\rightarrow$	$H = A\tilde{A} = AA_{\bar{2},\bar{3}}$	$H' = A\tilde{\tilde{A}} = AA_{\bar{1},\bar{2},\bar{5}}$
13	$H(H(H_{\bar{1},\bar{4}}H)_{\bar{5}})_{\bar{1},\bar{4}}$ $H(H(HH_{\bar{4},\bar{5}})_{\bar{1}})_{\bar{4},\bar{5}}$ $HH_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H)_{\bar{4}}$	$H(H(HH_{\bar{1},\bar{4}})_{\bar{5}})_{\bar{1},\bar{4}}$ $HH_{\bar{1},\bar{4}}(H_{\bar{1},\bar{4}}H)_{\bar{5}}$ $HH_{\bar{4},\bar{5}}(H_{\bar{4},\bar{5}}H)_{\bar{1}}$	$H(H(HH_{\bar{1},\bar{5}})_{\bar{4}})_{\bar{1},\bar{5}}$ $HH_{\bar{1},\bar{4}}(HH_{\bar{1},\bar{4}})_{\bar{5}}$
14	$H(H_{\bar{1},\bar{4}}HH_{\bar{1},\bar{5}})_{\bar{4},\bar{5}}$ $H(H_{\bar{1},\bar{5}}H_{\bar{4},\bar{5}}H)_{\bar{1},\bar{4}}$ $H(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{5}}H)_{\bar{1},\bar{4}}$ $H(H(H_{\bar{4},\bar{5}}H)_{\bar{1},\bar{3}})_{\bar{4},\bar{5}}$ $H(HH_{\bar{1},\bar{5}}H_{\bar{4},\bar{5}})_{\bar{1},\bar{4}}$ $HH_{\bar{1},\bar{4}}H_{\bar{4},\bar{5}}H_{\bar{1},\bar{5}}$ $HH_{\bar{4},\bar{5}}(HH_{\bar{4},\bar{5}})_{\bar{1},\bar{3}}$	$H(H_{\bar{1},\bar{4}}HH_{\bar{4},\bar{5}})_{\bar{1},\bar{5}}$ $H(H_{\bar{1},\bar{5}}HH_{\bar{1},\bar{4}})_{\bar{4},\bar{5}}$ $H(H_{\bar{4},\bar{5}}HH_{\bar{1},\bar{4}})_{\bar{1},\bar{5}}$ $H(HH_{\bar{1},\bar{4}}H_{\bar{1},\bar{5}})_{\bar{4},\bar{5}}$ $H(HH_{\bar{4},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{1},\bar{4}}$ $HH_{\bar{1},\bar{5}}(HH_{\bar{1},\bar{5}})_{\bar{3},\bar{4}}$ $HH_{\bar{4},\bar{5}}H_{\bar{1},\bar{5}}H_{\bar{1},\bar{4}}$	$H(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{4}}H)_{\bar{4},\bar{5}}$ $H(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}}H)_{\bar{1},\bar{5}}$ $H(H(H_{\bar{1},\bar{5}}H)_{\bar{3},\bar{4}})_{\bar{1},\bar{5}}$ $HH_{\bar{1},\bar{4}}H_{\bar{4},\bar{5}})_{\bar{1},\bar{5}}$ $HH_{\bar{1},\bar{4}}H_{\bar{1},\bar{5}}H_{\bar{4},\bar{5}}$ $HH_{\bar{1},\bar{5}}H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}}$
15	$H(H_{\bar{1},\bar{4}}(H_{\bar{1},\bar{5}}H)_{\bar{3}})_{\bar{4},\bar{5}}$ $H(H_{\bar{4},\bar{5}}(HH_{\bar{1},\bar{5}})_{\bar{3}})_{\bar{1},\bar{4}}$ $HH_{\bar{1},\bar{5}}(H_{\bar{1},\bar{4}}H_{\bar{4},\bar{5}})_{\bar{3}}$	$H(H_{\bar{1},\bar{4}}(H_{\bar{4},\bar{5}}H)_{\bar{3}})_{\bar{1},\bar{5}}$ $H(H(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{3}})_{\bar{4},\bar{5}}$ $HH_{\bar{4},\bar{5}}(H_{\bar{1},\bar{4}}H_{\bar{1},\bar{5}})_{\bar{3}}$	$H(H_{\bar{1},\bar{5}}(HH_{\bar{4},\bar{5}})_{\bar{3}})_{\bar{1},\bar{4}}$ $H(H(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{3}})_{\bar{1},\bar{5}}$

Inverses in dimension $n = p + q = 5$

Inverse formulas are not unique for $n > 4$ (continued)

$N \downarrow$	Abbreviation $\rightarrow$	$H = A\tilde{A} = AA_{\bar{2},\bar{3}}$	$H' = A\tilde{\tilde{A}} = AA_{\bar{1},\bar{2},\bar{5}}$
17	$H(H_{\bar{1},\bar{4}}(H_{\bar{1},\bar{4}}H_{\bar{1},\bar{5}})_{\bar{1}})_{\bar{1},\bar{5}}$ $H(H_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H_{\bar{4},\bar{5}})_{\bar{5}})_{\bar{4},\bar{5}}$ $H(H_{\bar{4},\bar{5}}(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{4}})_{\bar{1},\bar{4}}$	$H(H_{\bar{1},\bar{4}}(H_{\bar{1},\bar{4}}H_{\bar{4},\bar{5}})_{\bar{4}})_{\bar{4},\bar{5}}$ $H(H_{\bar{1},\bar{5}}(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{5}})_{\bar{4},\bar{5}}$ $H(H_{\bar{4},\bar{5}}(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{5}})_{\bar{1},\bar{5}}$	$H(H_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{1}})_{\bar{1},\bar{4}}$ $H(H_{\bar{4},\bar{5}}(H_{\bar{1},\bar{5}}H_{\bar{4},\bar{5}})_{\bar{5}})_{\bar{1},\bar{5}}$
18	$H(H_{\bar{1},\bar{4}}(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{1},\bar{3}})_{\bar{1},\bar{5}}$ $H(H_{\bar{4},\bar{5}}(H_{\bar{1},\bar{4}}H_{\bar{4},\bar{5}})_{\bar{3},\bar{4}})_{\bar{1},\bar{4}}$	$H(H_{\bar{1},\bar{4}}(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{3},\bar{4}})_{\bar{4},\bar{5}}$	$H(H_{\bar{1},\bar{5}}(H_{\bar{1},\bar{4}}H_{\bar{1},\bar{5}})_{\bar{1},\bar{3}})_{\bar{1},\bar{4}}$
15	$H'(H'(H'H'_3)_{\bar{4}})_{\bar{3}}$	$H'H'_3(H'_3H')_{\bar{4}}$	
16	$H'(H'(H'_3H')_{\bar{1},\bar{4}})_{\bar{3}}$	$H'(H'_3(H'H'_3)_{\bar{1},\bar{4}})$	

Alternative formulas for MV determinant norm for GAs of spaces of dimension $n = 5$ listed according to the number of negations. For example, the number of negations N in the first line is determined by four $H = AA_{\bar{2},\bar{3}}$, each including 2 negations, $(\bar{2}, \bar{3})$, and five explicit negations $(\bar{1}, \bar{4}) + \bar{5} + (\bar{1}, \bar{4})$ in the formula, resulting in $2 * 4 + 5 = 13$ total negations.

Mathematica example

Inverse in dimension $n = p + q = 6$

Why a single self-negated product does not work in $p + q = 6$ case?

- The culprit appear to be the **grade-4** blades. They cannot be eliminated by single self-product!
- For $n = 6$ we can simultaneously eliminate either grades 1, 2, 5 and 6 (then 0, 3 and 4 grades survive) or, alternatively, the grades 2, 3 and 6 (then grades 0, 1, 4 and 5 persist).
- There exists a subalgebra the base of which is formed by elements $\{1, \mathbf{e}_{1256}, \mathbf{e}_{1346}, \mathbf{e}_{2345}\}$ such that any self product of multivector

$$B = a_0 + a_{47}\mathbf{e}_{1256} + a_{49}\mathbf{e}_{1346} + a_{52}\mathbf{e}_{2345}$$

by grade-negated B will yield new MV having at least one grade-4 base element present.

- There is **only one** such an "non-eliminable" grade part in $p + q = 6$ case.

We will start by trying to eliminate this special grade-4 part.

Inverse in dimension $n = p + q = 6$

Pattern of linear combination

- Because matrix determinant of grade-4 MV is a square, just 4 MV product is required.
- Form prototype of a geometric product of the form

$$s_{4+4} = s_{4f} + s_{4g} =$$

$$b_1 B f_5 \left(f_4(B) f_3 \left(f_2(B) f_1(B) \right) \right) + b_2 B g_5 \left(g_4(B) g_3 \left(g_2(B) g_1(B) \right) \right),$$

where f_j, g_j denotes yet undetermined grade-4 negations and b_1, b_2 are unknown scalar coefficients of the linear combination.

Calculate explicit symbolic form of square root of the determinant of matrix representation of the multivector

$B = a_0 + a_{47}\mathbf{e}_{1256} + a_{49}\mathbf{e}_{1346} + a_{52}\mathbf{e}_{2345}$ and compare it with all possible negations of grade-4 substituted for f_j and g_j .

Inverse in dimension $n = p + q = 6$

Modelling the grade negation

- Multiply the grade-4 part by unknown coefficients p_{4jk} , where the index j denotes the negation number f_j, g_j in the self-product and k is the term in linear combination (in $p + q = 6$ case $k = 1$ for f and $k = 2$ for g).
- The coefficients p_{ijk} acquire the values ± 1 only, where -1 means that negation that changes sign of grade i is applied and $+1$ means the identity.
- The answer: $\{b_1 = -2/3, b_2 = -1/3, p_{411} = -1, p_{412} = 1, p_{421} = -1, p_{422} = 1, p_{431} = -1, p_{432} = -1, p_{441} = -1, p_{442} = 1, p_{451} = -1, p_{452} = 1\}$ and $\{b_1 = -1/3, b_2 = -2/3, p_{411} = 1, p_{412} = -1, p_{421} = 1, p_{422} = -1, p_{431} = -1, p_{432} = -1, p_{441} = 1, p_{442} = -1, p_{451} = 1, p_{452} = -1\}$.
- Check the answer with the most general **scalar + grade-4** multivector.

Inverse in dimension $n = p + q = 6$

The full determinant norm

- How to double the number of multivectors?
- Eliminate simultaneously either grades 1, 2, 5 and 6 (then grades 0, 3 and 4 remain), or, alternatively the grades 2, 3 and 6 (then grades 0, 1, 4 and 5 remain). All in all there are $\frac{6!}{2!4!} + \frac{6!}{3!3!} + 1 = 36$ elements of grades 2, 3 and 6 which can be eliminated simultaneously. This is more than 28 total elements of grades 1, 2, 5 and 6 that need to be eliminated.
- Therefore, if we replace B in s_{4+4} pattern by self-negated product $AA_{\bar{2},\bar{3},\bar{6}}$ we will have eight required terms.
- Then we find the other involutions using the multivector $a_0 + a_1\mathbf{e}_1 + a_{22}\mathbf{e}_{123} + a_{47}\mathbf{e}_{1256}$ and check result with general MV.
- Repeat the procedure for search of grade negations that consecutively eliminate all other grades.

Many solutions.

Determinant norm formulas for $n = p + q = 6$

N	Determinant norm for $n = 6$	$H = A\tilde{A} = AA_{\bar{2},\bar{3},\bar{6}}$
38	$\frac{1}{3}H(H(HH_{\bar{1},\bar{5}})_{\bar{4}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{1},\bar{4}}(H_{\bar{1},\bar{4}}H_{\bar{4},\bar{5}})_{\bar{4}})_{\bar{4},\bar{5}}$ $\frac{1}{3}HH_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H)_{\bar{4}} + \frac{2}{3}H(H_{\bar{1},\bar{4}}(H_{\bar{1},\bar{4}}H_{\bar{4},\bar{5}})_{\bar{4}})_{\bar{4},\bar{5}}$ $\frac{1}{3}H(H(HH_{\bar{1},\bar{5}})_{\bar{4}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{4},\bar{5}}(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{4}})_{\bar{1},\bar{4}}$ $\frac{1}{3}HH_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H)_{\bar{4}} + \frac{2}{3}H(H_{\bar{4},\bar{5}}(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{4}})_{\bar{1},\bar{4}}$	
34	$\frac{1}{3}HH(HH)_{\bar{1},\bar{4},\bar{5}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}}$	
36	$\frac{1}{3}HH(HH)_{\bar{1},\bar{4},\bar{5}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{4}})_{\bar{4}}$ $\frac{1}{3}HH(HH)_{\bar{1},\bar{4},\bar{5}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{4}}H_{\bar{4}})_{\bar{4}})_{\bar{1},\bar{4},\bar{5}}$ $\frac{1}{3}H(H_{\bar{1},\bar{5}}(HH)_{\bar{4}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}}$ $\frac{1}{3}HH(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{4}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}}$	

The integer N counts the total number of negations.

Determinant norm formulas for $n = p + q = 6$ (continued)

N	Determinant norm for $n = 6$	$H = A\tilde{A} = AA_{\bar{2},\bar{3},\bar{6}}$
38	$\frac{1}{3}H(H_{\bar{1},\bar{5}}(HH)_{\bar{4}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{4}})_{\bar{4}}$ $\frac{1}{3}HH(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{4}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{4}})_{\bar{4}}$ $\frac{1}{3}H(H_{\bar{1},\bar{5}}(HH)_{\bar{4}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{4}}H_{\bar{4}})_{\bar{4}})_{\bar{1},\bar{4},\bar{5}}$ $\frac{1}{3}HH(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{4}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{4}}H_{\bar{4}})_{\bar{4}})_{\bar{1},\bar{4},\bar{5}}$	
42	$\frac{1}{3}HH(HH)_{\bar{1},\bar{4},\bar{5}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}}$ $\frac{1}{3}H(H_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}}$	
44	$\frac{1}{3}H(H_{\bar{1},\bar{5}}(HH)_{\bar{4}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}}$ $\frac{1}{3}HH(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{4}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}}$ $\frac{1}{3}H(H_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{4}})_{\bar{4}}$ $\frac{1}{3}H(H_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{4}}H_{\bar{4}})_{\bar{4}})_{\bar{1},\bar{4},\bar{5}}$	
50	$\frac{1}{3}H(H_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}}$	

Besides these $16 + 4 = 20$ two term linear combinations one can find 72 formulas using three term combinations. **Example.**

Inverse formulas for $n = p + q \leq 6$

$Cl_{p,q}$	A^{-1}	
$p + q = 0$	$\frac{B}{AB},$	$B = 1$
$p + q = 1$	$\frac{B(AB)_{\bar{1}}}{AB(AB)_{\bar{1}}} = \frac{\widehat{A}}{AA},$	$B = 1$
$p + q = 2$	$\frac{C}{AC} = \frac{\widetilde{\widehat{A}}}{AA},$	$C = A_{\bar{1},\bar{2}}$
$p + q = 3$	$\frac{C(AC)_{\bar{3}}}{AC(AC)_{\bar{3}}} = \frac{\widetilde{\widetilde{\widehat{AAA}}}}{AAAA},$	$C = A_{\bar{1},\bar{2}}$
$p + q = 4$	$\frac{D}{AD},$	$D = A_{\bar{2},\bar{3}}(AA_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}$ or $D = A_{\bar{1},\bar{2}}(AA_{\bar{1},\bar{2}})_{\bar{3},\bar{4}}$
$p + q = 5$	$\frac{D(AD)_{\bar{5}}}{AD(AD)_{\bar{5}}},$	$D = A_{\bar{2},\bar{3}}(AA_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}$
$p + q = 6$	$\frac{G}{AG}, \quad G = \frac{1}{3}A_{\bar{2},\bar{3},\bar{6}}\left(H(HH)_{\bar{1},\bar{4},\bar{5}} + 2(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}}\right)$ or $G = \frac{1}{3}A_{\bar{2},\bar{3},\bar{6}}\left((H(HH)_{\bar{1},\bar{5}})_{\bar{4}}\right)_{\bar{1},\bar{5}} + 2(H_{\bar{4},\bar{5}}(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{4}})_{\bar{1},\bar{4}}$ with $H = AA_{\bar{2},\bar{3},\bar{6}} = \widetilde{AA}$	

Summary of formulas for inverse multivectors A^{-1} in Clifford algebras of dimension $n \leq 6$.

- 1 Introduction to geometric algebra
- 2 $Cl_{3,0}$ quantum mechanics
 - Standard quantum mechanics
 - How to dispense with matrices?
 - Two level model
 - Example of monolayer graphene
 - The algorithm
- 3 Multivector inverse for $n = p + q \leq 6$
 - Introduction
 - Grade-negated self-product
 - Inverse of general MV of algebras of dimension $n = p + q \leq 5$
 - Inverse of general MV of algebras of dimension $n = p + q = 6$
- 4 Coordinate-free solution of quaternionic Sylvester equation
- 5 Conclusions

Quaternionic Sylvester equation (non-commutative)

Solution steps of the quaternionic equation $ax + xb = c$

- 1 Multiply the equation from left by a to get

$$a^2x + axb = ac.$$

- 2 Then multiply from right by conjugate quaternion $\bar{b}$ to get

$$ax\bar{b} + xb\bar{b} = c\bar{b}.$$

- 3 Since both equations are linear we can add,

$$(a^2x + xb\bar{b}) + (axb + ax\bar{b}) = ac + c\bar{b}.$$

- 4 Since $b\bar{b}$ and $b + \bar{b}$ are scalars, we can write

$$(a^2x + b\bar{b}x) + a(b + \bar{b})x = ac + c\bar{b},$$

from which we have the solution:

$$x = \left[(a^2 + b\bar{b}) + a(b + \bar{b}) \right]^{-1} (ac + c\bar{b}).$$

Clifford algebra Sylvester equation

Why we can solve quaternionic case ?

It was possible to form **two scalars**, $b + \bar{b}$ and $b\bar{b}$, that commute with all quaternions!

The $Cl_{p,q}$ algebra has a center, which commute with all elements of the considered algebra,

$$\text{Cen}(\mathcal{A}) = \{s \in \mathcal{A} \mid as = sa, \forall a \in \mathcal{A}\}.$$

The requirement for the quaternion method is that $(a + \bar{a}) = \text{Cen}_1(Cl_{p,q})$ **and** $(a\bar{a}) = \text{Cen}_2(Cl_{p,q})$.

Which $Cl_{p,q}$ algebras satisfy these requirements?

Clifford algebra Sylvester equation

$Cl_{p,q}$	$\bar{a}$	$\text{Cen}_1 = a + \bar{a}$	$\text{Cen}_2 = a\bar{a}$
$Cl_{2,0}$	$a_{\bar{1},\bar{2}}$	$2a_0$	$a_0^2 - a_1^2 - a_2^2 + a_3^2$
$Cl_{1,1}$	$a_{\bar{1},\bar{2}}$	$2a_0$	$a_0^2 - a_1^2 + a_2^2 - a_3^2$
$Cl_{0,2}$	$a_{\bar{1},\bar{2}}$	$2a_0$	$a_0^2 + a_1^2 + a_2^2 + a_3^2$
$Cl_{3,0}$	$a_{\bar{1},\bar{2}}$	$2(a_0 + a_{123}I_3)$	$a_0^2 - a_1^2 - a_2^2 - a_3^2 + a_{12}^2 + a_{13}^2 + a_{23}^2 - a_{123}^2$ $+ 2(a_0a_{123} - a_1a_{23} + a_2a_{13} - a_3a_{12})I_3$
$Cl_{2,1}$	$a_{\bar{1},\bar{2}}$	$2(a_0 + a_{123}I_3)$	$a_0^2 - a_1^2 - a_2^2 + a_3^2 + a_{12}^2 - a_{13}^2 - a_{23}^2 + a_{123}^2$ $+ 2(a_0a_{123} - a_1a_{23} + a_2a_{13} - a_3a_{12})I_3$
$Cl_{1,2}$	$a_{\bar{1},\bar{2}}$	$2(a_0 + a_{123}I_3)$	$a_0^2 - a_1^2 + a_2^2 + a_3^2 - a_{12}^2 - a_{13}^2 + a_{23}^2 - a_{123}^2$ $+ 2(a_0a_{123} - a_1a_{23} + a_2a_{13} - a_3a_{12})I_3$
$Cl_{0,3}$	$a_{\bar{1},\bar{2}}$	$2(a_0 + a_{123}I_3)$	$a_0^2 + a_1^2 + a_2^2 + a_3^2 + a_{12}^2 + a_{13}^2 + a_{23}^2 + a_{123}^2$ $+ 2(a_0a_{123} - a_1a_{23} + a_2a_{13} - a_3a_{12})I_3$

Clifford algebras that allow coordinate-free solution of equation in the form. The last two columns represent CA centers. I_3 is the corresponding pseudoscalar. **Example.**

- 1 Introduction to geometric algebra
- 2 $Cl_{3,0}$ quantum mechanics
 - Standard quantum mechanics
 - How to dispense with matrices?
 - Two level model
 - Example of monolayer graphene
 - The algorithm
- 3 Multivector inverse for $n = p + q \leq 6$
 - Introduction
 - Grade-negated self-product
 - Inverse of general MV of algebras of dimension $n = p + q \leq 5$
 - Inverse of general MV of algebras of dimension $n = p + q = 6$
- 4 Coordinate-free solution of quaternionic Sylvester equation
- 5 Conclusions

Conclusions

- The GA can be applied to solve eigenvalue problems. The answers found in this way appear to be rather compact and may be interpreted geometrically. Eigen-vector concept generalization to eigen-blades (eigen-multivectors?).
- Inverse of GA multivector can be written as a sum of self-negated product. The number of factors in the product is determined by 8-periodicity table. The formula for $n = 6$ alone produces determinant norms and inverses of all $Cl_{p,q}$ algebras of $n \leq 6$ (onion like structure).
- Sylvester equation can be solved in a coordinate free way for $Cl_{p,q}$ algebra which has two centers, i. e. for $n \leq 3$.

Conclusions

- The GA can be applied to solve eigenvalue problems. The answers found in this way appear to be rather compact and may be interpreted geometrically. Eigen-vector concept generalization to eigen-blades (eigen-multivectors?).
- Inverse of GA multivector can be written as a sum of self-negated product. The number of factors in the product is determined by 8-periodicity table. The formula for $n = 6$ alone produces determinant norms and inverses of all $Cl_{p,q}$ algebras of $n \leq 6$ (onion like structure).
- Sylvester equation can be solved in a coordinate free way for $Cl_{p,q}$ algebra which has two centers, i. e. for $n \leq 3$.

Conclusions

- The GA can be applied to solve eigenvalue problems. The answers found in this way appear to be rather compact and may be interpreted geometrically. Eigen-vector concept generalization to eigen-blades (eigen-multivectors?).
- Inverse of GA multivector can be written as a sum of self-negated product. The number of factors in the product is determined by 8-periodicity table. The formula for $n = 6$ alone produces determinant norms and inverses of all $Cl_{p,q}$ algebras of $n \leq 6$ (onion like structure).
- Sylvester equation can be solved in a coordinate free way for $Cl_{p,q}$ algebra which has two centers, i. e. for $n \leq 3$.

Bibliography



A. Dargys and A. Acus,

Calculation of Quantum Eigens with Geometrical Algebra Rotors

Advances in Applied Clifford Algebras 27 (1), 2017, p. 241-253.



A. Acus and A. Dargys,

The inverse of a multivector: beyond the threshold $p + q = 5$

Advances in Applied Clifford Algebras (2018) 28:65.



A. Dargys and A. Acus,

Solutions of $ax + bx = c$ in Clifford algebras

submitted to Advances in Applied Clifford Algebras.

Thanks!